

Case Study Report

Advantages of Using AI in Government and Public Sector

A Case Study of AI-Driven Tax Fraud Detection and a Proposal for Predictive Water-Infrastructure Maintenance

Imran Khan

ITAI 2372

Professor Ganti

Houston City College

Date: July 11, 2026

1. Introduction

Governments across the world are under constant pressure to deliver more effective services with constrained budgets, ageing infrastructure, and rising citizen expectations. Artificial Intelligence (AI) — encompassing machine learning, natural language processing, computer vision, and large-scale data analytics — has emerged as a practical tool for closing this gap. Public administrations are increasingly using AI to detect fraud, forecast infrastructure failures, coordinate disaster response, personalize citizen services, and support evidence-based policymaking.

Fraud detection is one of the clearest examples of this shift, because it combines two conditions where AI performs well: very large volumes of structured and unstructured data, and a repetitive pattern-recognition task that benefits from consistent, tireless analysis rather than case-by-case manual review. Governments handling billions of transactions, records, and citizen interactions each year are natural candidates for machine learning and analytics tools that can surface the small proportion of cases genuinely worth investigating.

This report examines one well-documented application of AI in the public sector — the United Kingdom's HM Revenue & Customs (HMRC) "Connect" system, which uses data analytics and, more recently, AI-assisted risk scoring to detect tax fraud and evasion. The case study is analyzed in terms of the problem it addresses, the technology deployed, the outcomes achieved, and the challenges that have surfaced. Building on the lessons of this case, the report then proposes an original AI application — a predictive-maintenance and leak-detection system for municipal water infrastructure — and evaluates its feasibility, expected benefits, and risks. The two examples are deliberately drawn from different domains, fiscal enforcement and physical infrastructure, to illustrate that the underlying AI pattern of continuous data monitoring paired with human-supervised decision-making generalizes well across very different government functions.

2. Case Study Analysis: HMRC's Connect System

2.1 Problem or Need Addressed

Tax authorities worldwide face a persistent "tax gap" — the difference between tax theoretically owed and tax collected — driven by evasion, avoidance, and error. Before large-scale data analytics, HMRC investigators relied heavily on manual cross-checking of tax returns against limited third-party records, a slow process that could only cover a small fraction of taxpayers and was ill-suited to detecting fraud hidden across multiple accounts, shell companies, or undeclared income streams.

2.2 AI Tools and Techniques Implemented

HMRC's response, launched from 2010 onward, is built around the Connect system, which aggregates and cross-references billions of data points from banks, employers, the Land Registry, flight records, and online marketplaces to flag inconsistencies between declared income and observed financial behavior.

Connect has since been layered with more advanced AI capabilities:

- Risk-scoring engines that automatically prioritize cases most likely to involve fraud, allowing human investigators to focus effort where it matters most.

- Entity resolution technology (developed with the analytics firm Quantexa) that links disparate records — company filings, addresses, transactions — to uncover complex fraud networks hidden behind shell companies.
- AI-assisted social media monitoring, used in criminal investigations, that scans public posts for lifestyle indicators (e.g., luxury holidays or physical activity) inconsistent with declared income or disability claims.

Crucially, HMRC has maintained a human-in-the-loop model: AI outputs are risk scores and flags, not final decisions, and compliance officers review findings before any enforcement action is taken.

2.3 Outcomes and Benefits Achieved

Independent analysis has found that HMRC's Connect analytics platform delivered a return on investment of approximately 37:1 in additional revenue relative to its running cost, one of the more striking documented figures for AI-enabled government systems. The system has also enabled targeted, lower-cost compliance campaigns — such as the Let Property Campaign for landlords and the "One to Many" letter program, which nudges taxpayers toward voluntary disclosure based on data-driven risk patterns rather than resource-intensive formal investigations. AI-assisted case triage has additionally allowed investigators to process far larger volumes of data than manual methods would allow, compressing the time between an anomaly being detected and a case being opened.

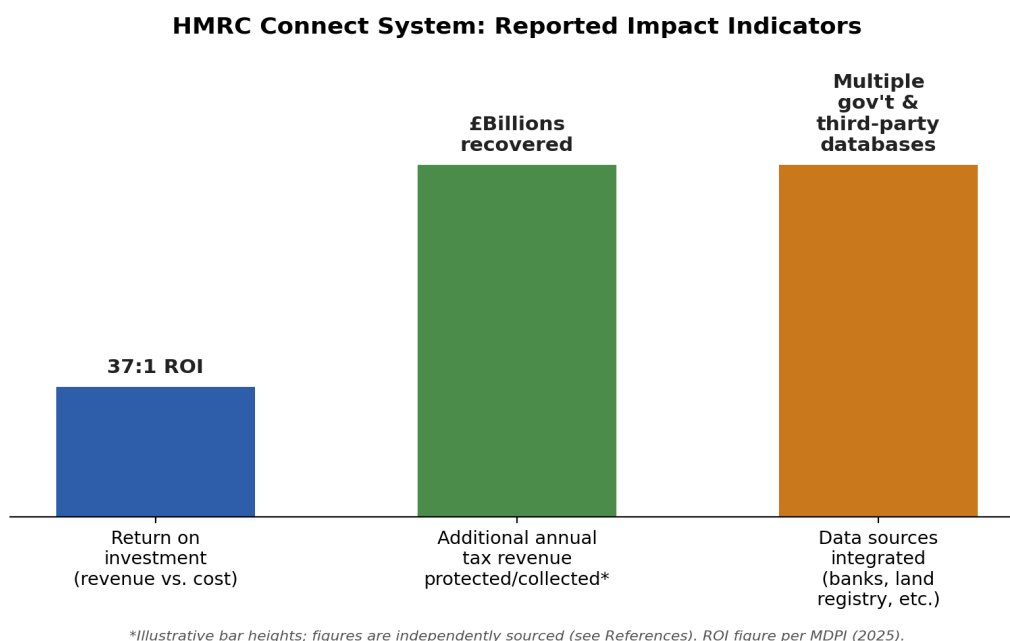


Figure 1. Summary of publicly reported impact indicators for HMRC's Connect system.

2.4 Challenges and Limitations Observed

- Transparency and disclosure: HMRC has been reluctant to disclose the scope of its AI use, and the Information Commissioner's Office initially backed that refusal; a tribunal later ordered greater disclosure, illustrating tension between operational secrecy and public accountability.
- Privacy concerns: automated monitoring of personal and social media data raises questions about proportionality and the boundaries of state surveillance, even when confined to criminal investigations.

- Algorithmic risk: legal commentary has flagged risks of algorithmic bias, opacity of reasoning, and weak independent oversight as AI is used more heavily in enforcement decisions.
- Continued reliance on human judgement: because flagged anomalies can have innocent explanations, human review remains essential, which limits how far the process can be fully automated without risking wrongful action against taxpayers.

2.5 Case Study Summary

Table 1 condenses the analysis above into the four evaluation dimensions used throughout this report.

Dimension	Summary	Illustrative Example
Problem addressed	Manual cross-checking could not keep pace with the scale and complexity of tax evasion.	Undeclared rental income; hidden offshore assets.
AI tools used	Big-data matching, AI risk scoring, entity resolution, social media monitoring.	Quantexa entity-resolution platform.
Outcomes achieved	Approx. 37:1 return on investment; faster, better-targeted campaigns.	Let Property Campaign; "One to Many" letters.
Challenges observed	Limited transparency, privacy concerns, and algorithmic accountability risks.	ICO/tribunal disclosure dispute (2025).

3. Innovative Proposal: AquaSense-AI — Predictive Maintenance for Municipal Water Infrastructure

3.1 The Governance Problem

Ageing water infrastructure is one of the least visible but most costly challenges facing municipal governments. Pipe bursts and undetected leaks waste treated water, disrupt roads and traffic, and are typically discovered only after a failure has already occurred, at which point repair costs and service disruption are far higher than they would be with early intervention. Most municipal water utilities still rely on scheduled inspections and citizen complaints rather than continuous, data-driven monitoring — a reactive model like the one tax authorities used before systems like Connect.

3.2 Proposed AI Application

AquaSense-AI would apply the same underlying logic as HMRC's Connect system — continuous data aggregation plus AI-driven anomaly detection with human-in-the-loop decision-making — to water infrastructure. The proposed workflow is illustrated in Figure 2.

Proposed AquaSense-AI Workflow: From Sensor Data to Field Action

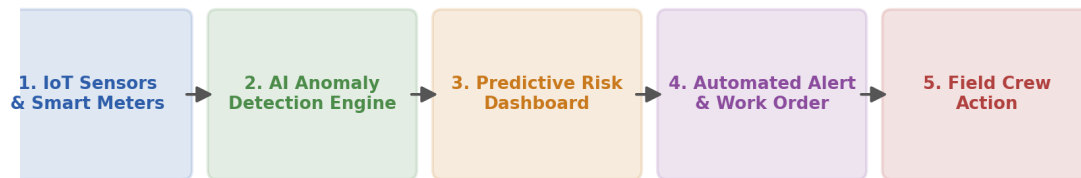


Figure 2. Proposed AquaSense-AI workflow from sensor data to field action.

- Data layer: acoustic sensors and smart meters installed at strategic points in the water network continuously stream flow, pressure, and vibration data.
- AI analysis layer: a machine-learning anomaly-detection model trained on historical failure data identifies pressure drops, unusual flow patterns, or acoustic signatures consistent with leaks or corrosion, ranking pipe segments by risk — mirroring Connect's risk-scoring approach.
- Decision layer: a dashboard presents prioritized risk areas to utility engineers, who review and confirm before a work order is issued, preserving human oversight over any action taken.
- Action layer: field crews receive targeted, pre-diagnosed repair or inspection tasks rather than searching for the source of a reported problem after the fact.

3.3 Justification and Expected Outcomes

This proposal is justified by the demonstrated success of analogous anomaly-detection systems in other public-sector domains, including tax administration. Expected benefits include earlier detection of leaks and reduced non-revenue water loss, lower emergency repair costs relative to reactive maintenance, reduced service disruption for residents, and better long-term capital planning, since utilities would gain a data-driven view of which infrastructure segments are deteriorating fastest. As with Connect, the system does not replace engineers; it directs their limited time toward the highest-risk locations first.

A phased rollout would make the proposal more financially realistic for municipalities of varying size. In Phase 1, sensors would be installed only on the highest-risk or highest-consequence segments of the network, such as pipes beneath major roads or near critical facilities, allowing the utility to prove the model's accuracy before wider investment. In Phase 2, coverage would expand to the broader network using a mix of fixed sensors and mobile acoustic surveys, with the AI model retrained continuously as more failure data becomes available. In Phase 3, the risk-scoring output could be integrated directly into the utility's capital-planning process, so that long-term pipe-replacement budgets are informed by predicted failure likelihood rather than pipe age alone. This staged approach mirrors how HMRC's own analytics capability grew incrementally over more than a decade rather than being deployed all at once.

3.4 Anticipated Challenges

- Upfront capital cost of sensor networks, which may be difficult to justify for smaller municipal budgets without phased rollout or grant funding.

- Data quality and integration challenges where legacy pipe-network records are incomplete or inconsistently digitized.
- Cybersecurity risk, since connected infrastructure sensors expand the attack surface for critical public utilities.
- Change management: utility staff need training and trust-building to act on AI-generated risk scores rather than only responding to citizen complaints.

4. Conclusion

The HMRC Connect case study demonstrates that AI, when combined with rich data integration and sustained human oversight, can deliver substantial, measurable returns in a public-sector function that was previously slow and resource constrained. It also shows that these gains come with real governance costs — particularly around transparency, privacy, and algorithmic accountability — that must be actively managed rather than assumed away. The proposed AquaSense-AI system applies the same core principle, continuous data monitoring paired with AI-prioritized human decision-making, to a different but equally pressing public problem: ageing water infrastructure. Taken together, the case study and the proposal suggest that the most durable public-sector AI applications are not those that remove humans from decision-making, but those that use AI to direct scarce human attention to where it is needed most.

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